

IT436 Cloud Computing

Cloud Networking

Agenda

- Introduction
- Traffic Characteristics
- Physical Structure

Introduction



coursera



amazon

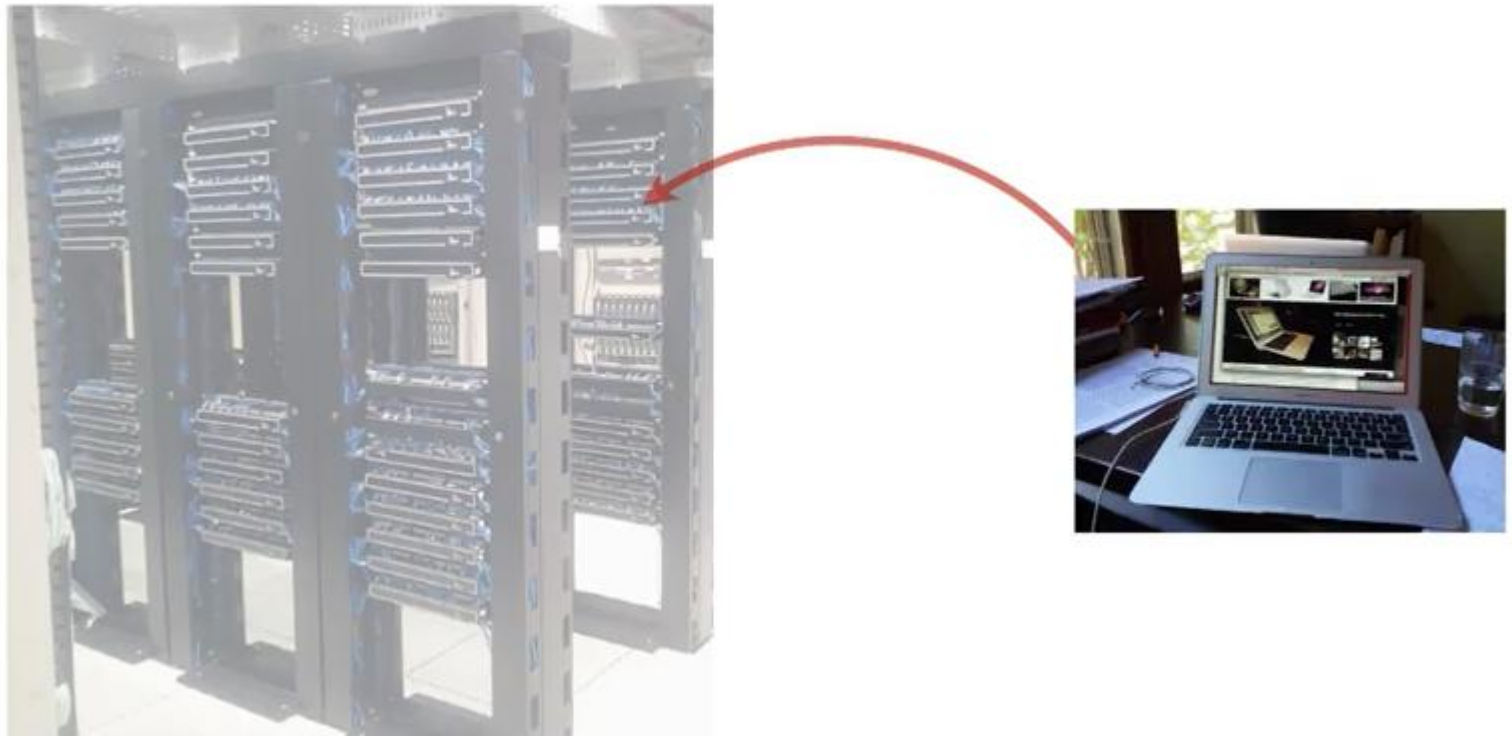
Google

bing



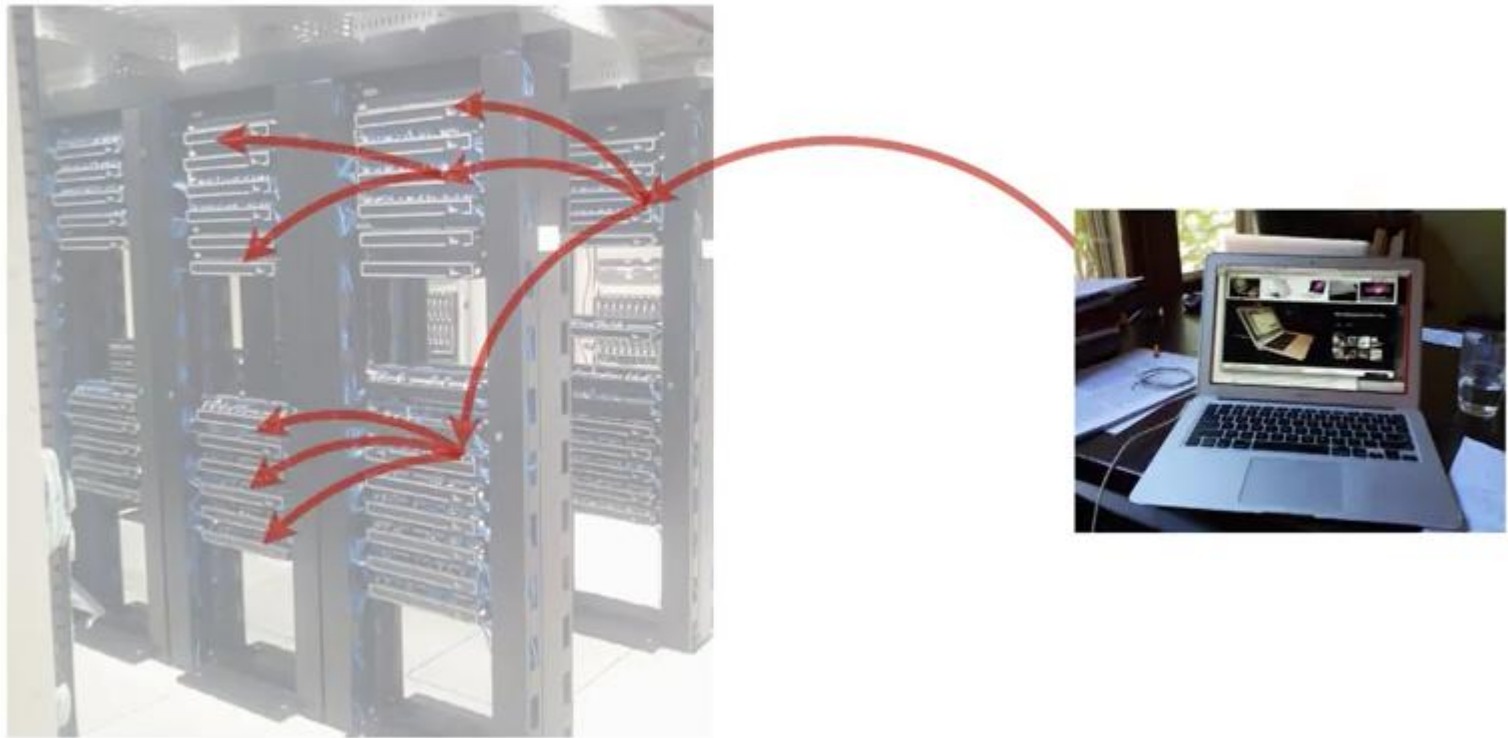
Introduction

Let's take a look at how web search works:



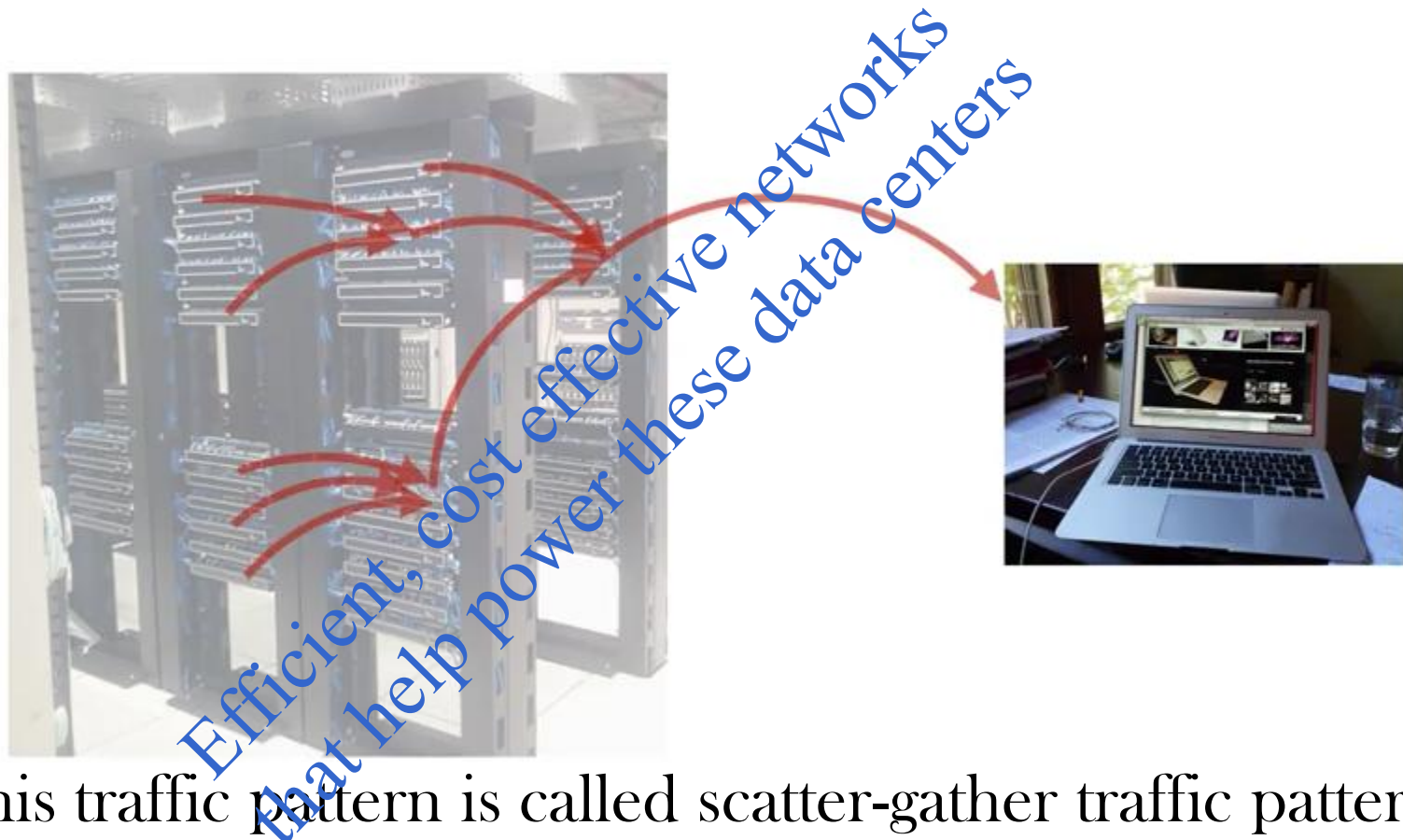
Introduction

Let's take a look at how web search works:



Introduction

Let's take a look at how web search works.



This traffic pattern is called scatter-gather traffic pattern

Introduction

In the remainder of the part, we will focus on:

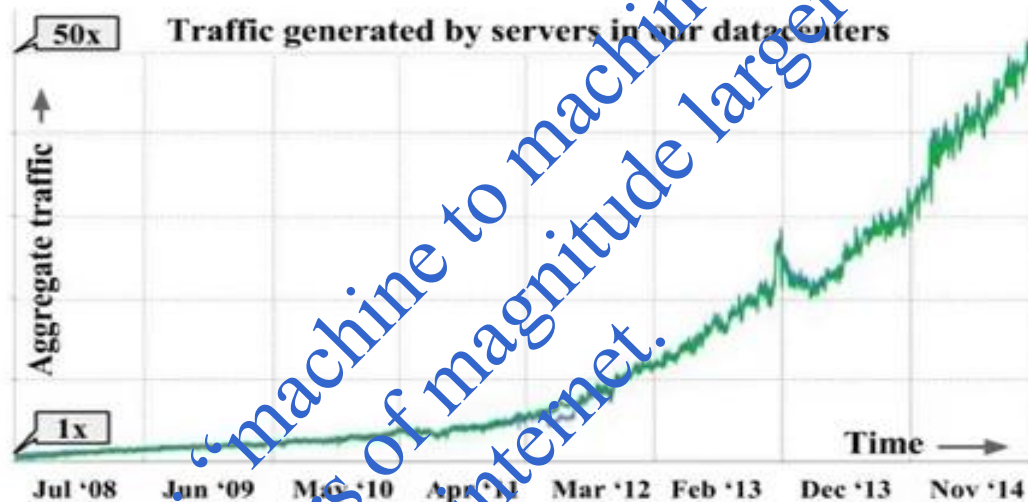
- Networking within the data center and not the carrier networks that deliver the information to and from the data center and your device.
- This includes network equipment, software, and standards used to create networks within large cloud data centers.



Traffic Characteristics

Traffic Characteristics

Traffic characteristics: growing volume



"Jupiter Rising: A Decade of Clos Topologies and Centralized Control in Google's Datacenter Network", Arjun Singh et al. @ **Google**, ACM SIGCOMM'15

Traffic Characteristics

Locality	Percentage
Rack	12.9
Cluster	57.5
DC	11.9
Inter-DC	17.7

Facebook

“Inside the Social Network’s (Datacenter) Network”
Arjun Roy et al., ACM SIGCOMM’15

Traffic Characteristics

■ Concurrent Flows

Facebook

“Inside the Social Network’s (Datacenter) Network”
Arjun Roy et al., ACM SIGCOMM’15

“Web servers and cache hosts have **100s**
to **1000s** of concurrent connections”

“Hadoop nodes have approximately **25**
concurrent connections on average.”

1500 server cluster @ ??

“The Nature of Datacenter Traffic: Measurements & Analysis”
Srikanth Kandula et al. (Microsoft Research), ACM IMC’09

“median numbers of correspondents for a
server are **two** (other) servers within its
rack and **four** servers outside the rack”

Traffic Characteristics

- Flows Arrival Rate

Facebook

“Inside the Social Network’s (Datacenter) Network”
Arjun Roy et al., ACM SIGCOMM’15

“median inter-arrival times of
approximately **2ms**”

1500 server cluster @ ??

“The Nature of Datacenter Traffic: Measurements & Analysis”
Srikanth Kandula et al. (**Microsoft** Research), ACM IMC’09

< 0.1x Facebook’s rate

Traffic Characteristics

■ Flows Size

Facebook

“Inside the Social Network’s (Datacenter) Network”
Arjun Roy et al., ACM SIGCOMM’15

Hadoop: median flow < 1KB
< 5% exceed 1MB or 100sec

Caching: most flows are long-lived
... but bursty internally

1500 server cluster @ ??

“The Nature of Datacenter Traffic: Measurements & Analysis”
Srikanth Kandula et al. (Microsoft Research), ACM IMC’09

> 80% of the flows last < 10sec
> 50% bytes are in flows lasting less < 25sec

Traffic Characteristics

- What does data center traffic look like?

It depends on applications, scale, network design.

Traffic Characteristics

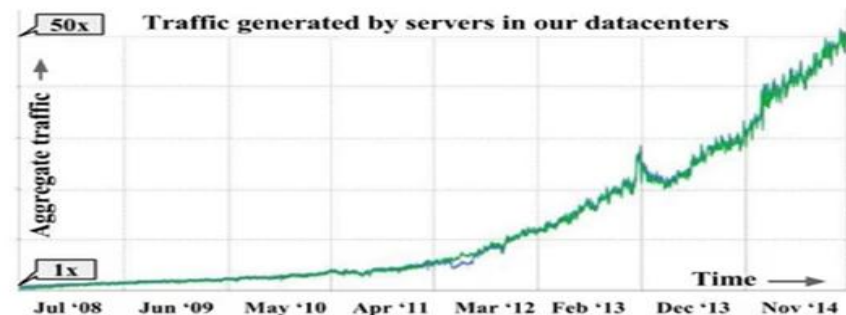
So, what are the implications for networking?

- Data center internal traffic is big.
 - Tight deadline for network I/O
 - Congestion and TCP incast.
-
- Need for isolation across applications
 - Centralized control at the flow level may be difficult.

Traffic Characteristics

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Traffic Characteristics

So, what are the implications for networking?

- Data center internal traffic is big.
- Tight deadline for network I/O

Suppose: server response-time is 10ms for 99% of requests; 1s for 1%

#Servers	Requests 1s or slower
1	1%
100	63%

- Congestion and TCP incast.
- Need for isolation across applications
- Centralized control at the flow level may be difficult.

Traffic Characteristics

“Cisco Global Cloud Index: Forecast and Methodology, 2012–2017”

- Data center traffic will grow by 25% each year at least through 2017.
- By 2017, over 2/3 of all data center traffic will be based in the cloud and 76% percent of this traffic will be between devices within the cloud data center.

Traffic Characteristics

“Cisco Global Cloud Index: Forecast and Methodology, 2012–2017”

- Server virtualization will have a large impact on cloud data center networks.



Physical Structure

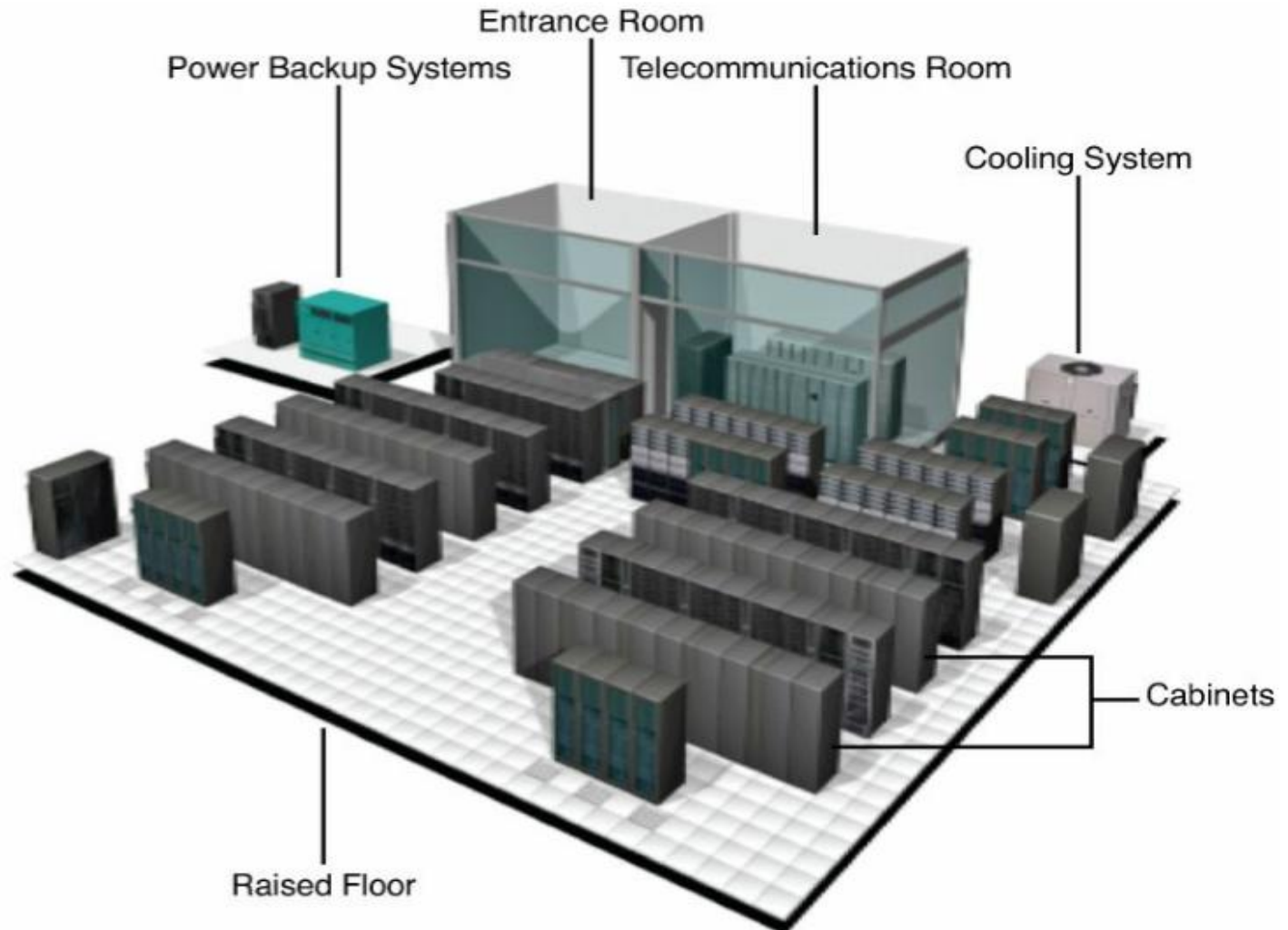
Physical Structure



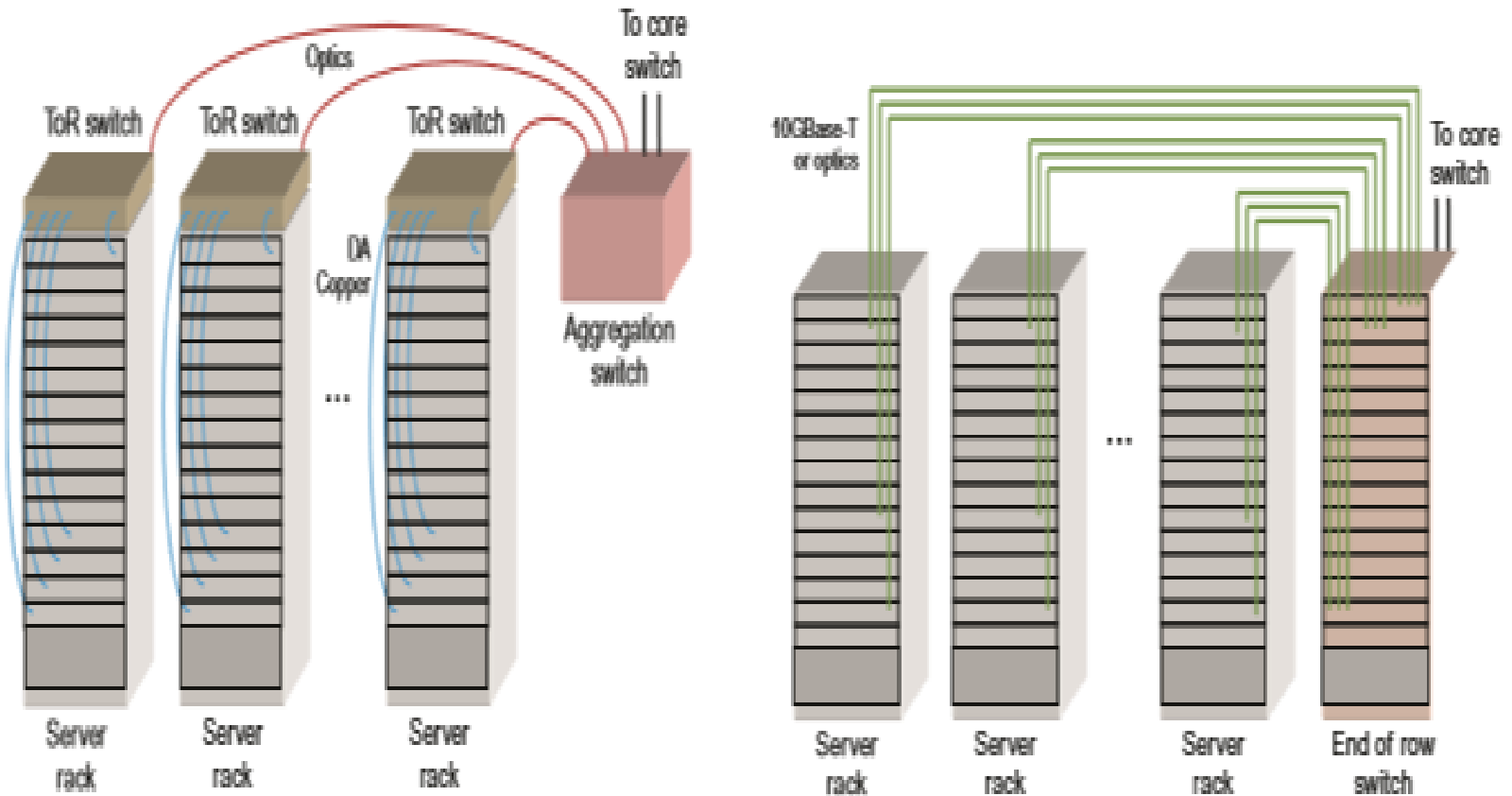
Data Center

- A facility to house, manage, and support computing resources that are considered critical for one or more organizations.
 - ✓ power backup structures
 - ✓ Cooling systems
 - ✓ Special-purpose rooms (e.g. entrance and telecommunications).
 - ✓ Equipment cabinets.
 - ✓ Structured cabling.
 - ✓ Network devices.
 - ✓ Storage systems, servers,

Data Center



Physical Network Layout Considerations



Physical Layout

Why ToR?!!

- High-performance servers and virtualization
- Supports east-west (server-to-server) traffic
- Copper stays “In Rack”. No large copper cabling infrastructure required.
- Lower cabling costs. Less infrastructure dedicated to cabling and patching.
- Cleaner cable management.
- Modular and flexible “per rack” architecture.
- Short copper cabling to servers allows for low power, low cost in the future.
- Ready for Unified Fabric today.

Physical Layout

Why **not** ToR?!!

- More switches to manage.
- Potential Interoperability Issues
- Maintenance, Equipment & Cabling Costs
- Switch Port Utilization
- Increased Power Consumption
- Cooling & Failure Rate Considerations
- More ports required in the aggregation.

Physical Layout

Why **not** ToR?!!

- Potential scalability concerns (STP Logical ports, aggregation switch density).
- More Layer 2 server-to-server traffic in the aggregation.
- Racks connected at Layer 2.
- More STP instances to manage.

Physical Layout

Why EoR?!!

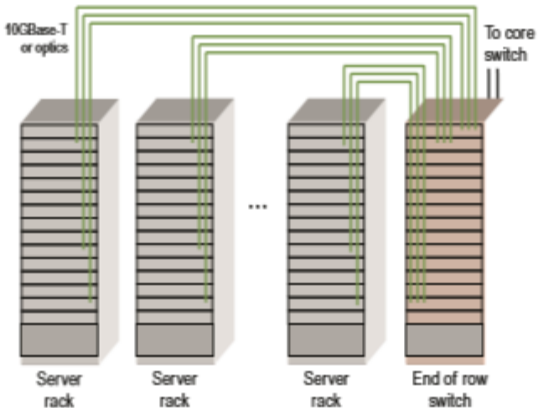
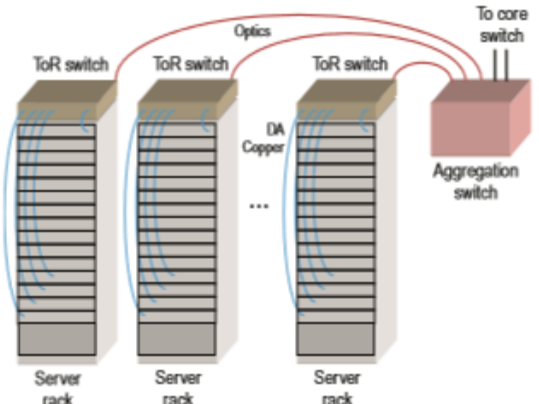
- Fewer switches to manage.
- Potentially lower switch costs, lower maintenance costs.
- Fewer ports required in the aggregation.
- Racks connected at Layer 1. Fewer STP instances to manage (per row, rather than per rack).

Physical Layout

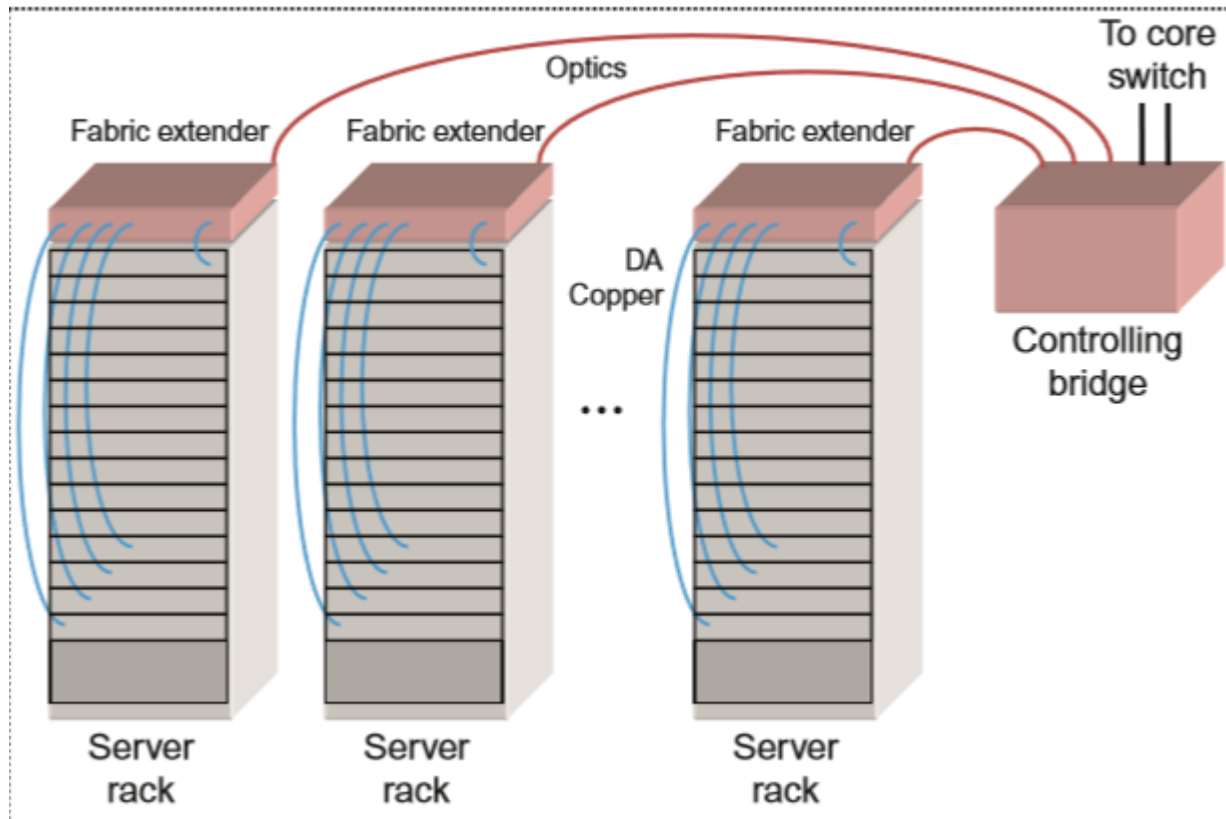
Why not EoR?!!

- Requires an expensive, bulky, rigid, copper cabling infrastructure. Fraught with cable management challenges.
- More infrastructure required for patching and cable management.
- Long twisted pair copper cabling limits the adoption of lower power higher speed server I/O.

Physical Network Layout Considerations

	Design	Description	Pros	Cons
	EoR End of Row	▶ All wiring is run from each server to the end of the row (last rack)	▶ Fewer switches ▶ Can accommodate mixture of 10G and high density 1G server environments	▶ High density of physical connections in EoR rack(s) ▶ More physical wiring components; patch panels, cable trays, etc.
	ToR Top of Rack	▶ All servers aggregate into switches located at the top of the individual server rack/cabinet.	▶ Greatly reduced wiring which will save DC build cost ▶ Troubleshooting will generally be easier ▶ Easy to migrate individual racks from 1G to 10G service	▶ More switches ▶ Not enough density to accommodate high density 1G environment

Physical Network Layout Considerations



Physical Layout

Why Fabric Extender?!!

- Fewer switches to manage.
- Fewer ports required in the aggregation area. (End of Row) Racks connected at Layer 1 via fiber, extending Layer 1 copper to servers in-rack. Fewer STP instances to manage. (End of Row)
- Unique control plane per hundreds of ports, lower skill set required for replacement. (End of Row)

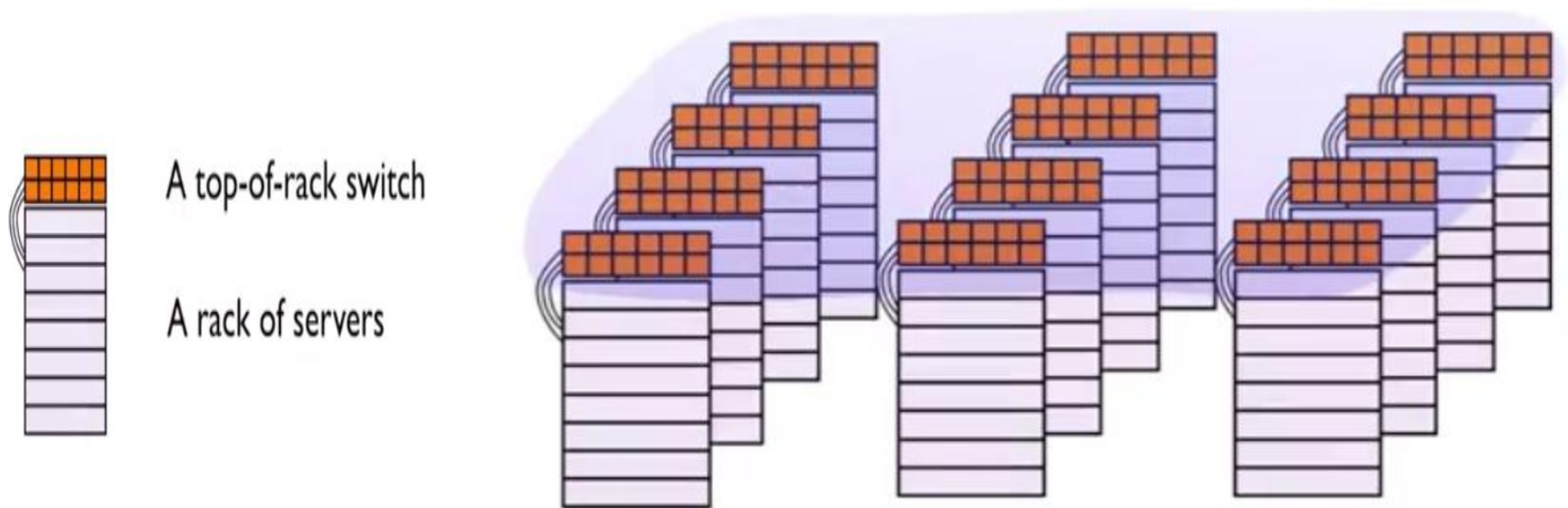
Physical Layout

Why Fabric Extender?!!

- Copper stays “In Rack”. No large copper cabling infrastructure required. (Top of Rack)
- Lower cabling costs. Less infrastructure dedicated to cabling and patching. Cleaner cable management. (Top of Rack)
- Modular and flexible “per rack” architecture.

Physical Structure

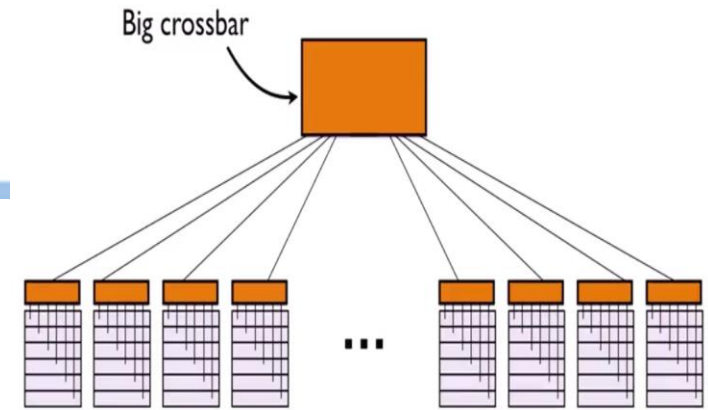
How to connect different racks of servers through a network?



Aggregation switches are typically used to connect a number of ToR switches to a core switch/router.

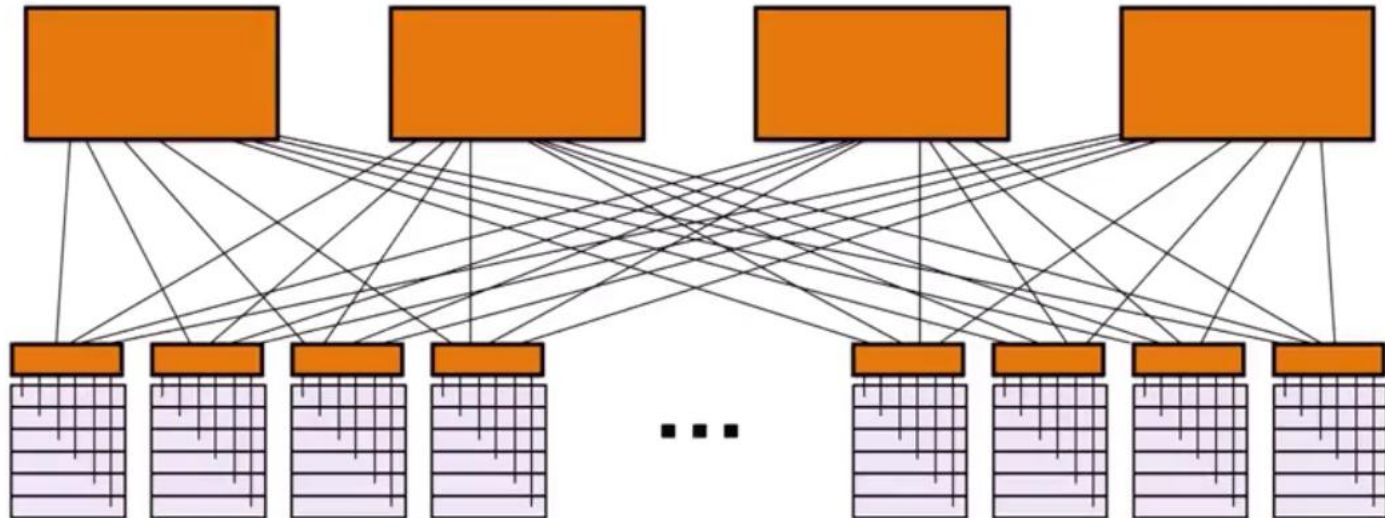
Physical Structure

Very expensive N^2



- A top of rack (ToR) switches containing a large number of 1Gb Ethernet (1GbE)
- The ToR switches are connected to each other through an Ethernet aggregation switch
- These ToR switch uplink ports to the aggregation switches traditionally use multiple 1GbE ports in a link aggregation group or 10GbE ports.
- As data centers grew larger, aggregation switches ran out of ports, requiring the use of large switch/routers that are typically used to interconnect multiple LANs on large corporate campuses and are similar in some ways to large IP routers used by internet service providers

Physical Structure



Physical Structure

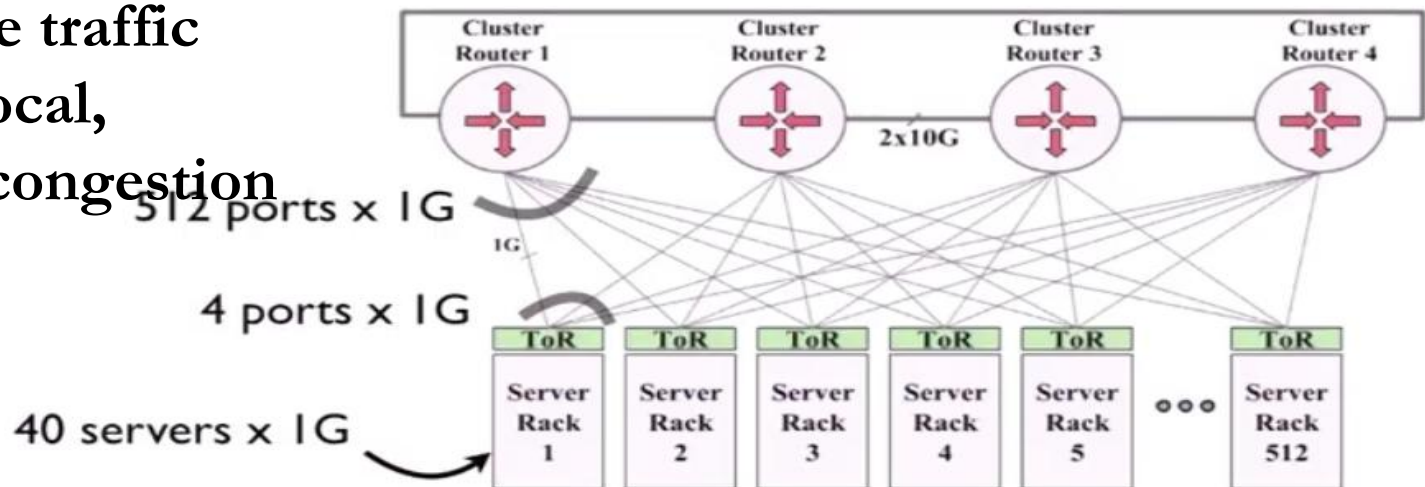
Example

ACM SIGCOMM, 2015

Jupiter Rising: A Decade of Clos Topologies and Centralized Control in Google's Datacenter Network

Arjun Singh, Joon Ong, Amit Agarwal, Glen Anderson, Ashby Armistead, Roy Bannon, Seb Boving, Gaurav Desai, Bob Felderman, Paulie Germano, Anand Kanagala, Jeff Provost, Jason Simmons, Eiichi Tanda, Jim Wanderer, Urs Hölzle, Stephen Stuart, and Amin Vahdat
Google, Inc.

- The top of the rack switch has 1/10 of the rack server speeds.
- Most of the traffic has to be local, otherwise congestion occurs



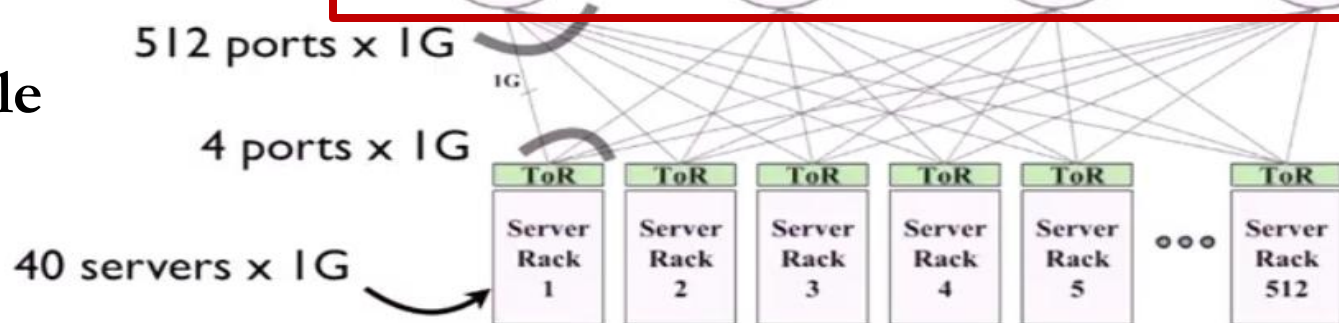
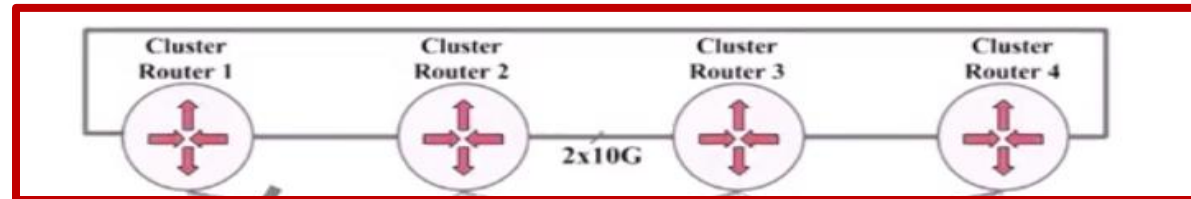
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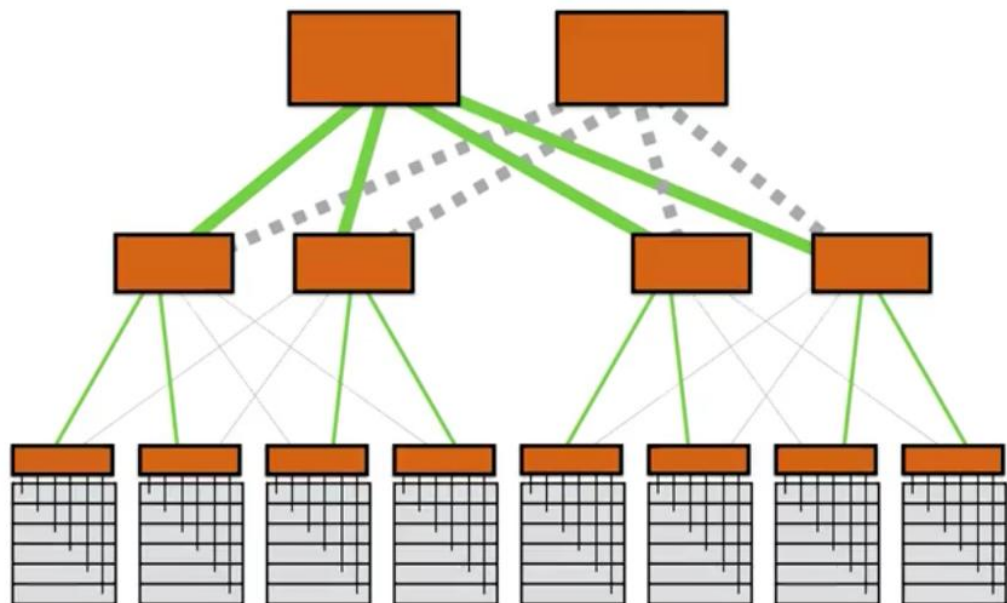


Physical Structure

Another Solution

- This way to can add more racks at the expenses of reduced capacity.
- Still congestion is possible at the top of the tree with increased machine-to-machine traffic
- Failure can partition the network

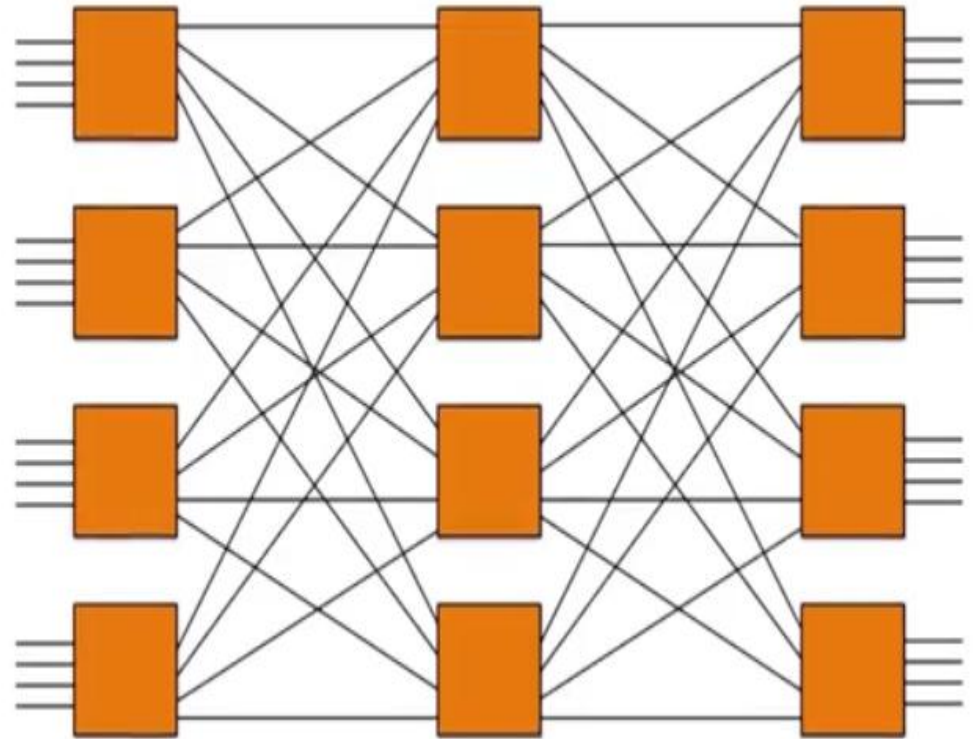
Alternative: tree network



Physical Structure

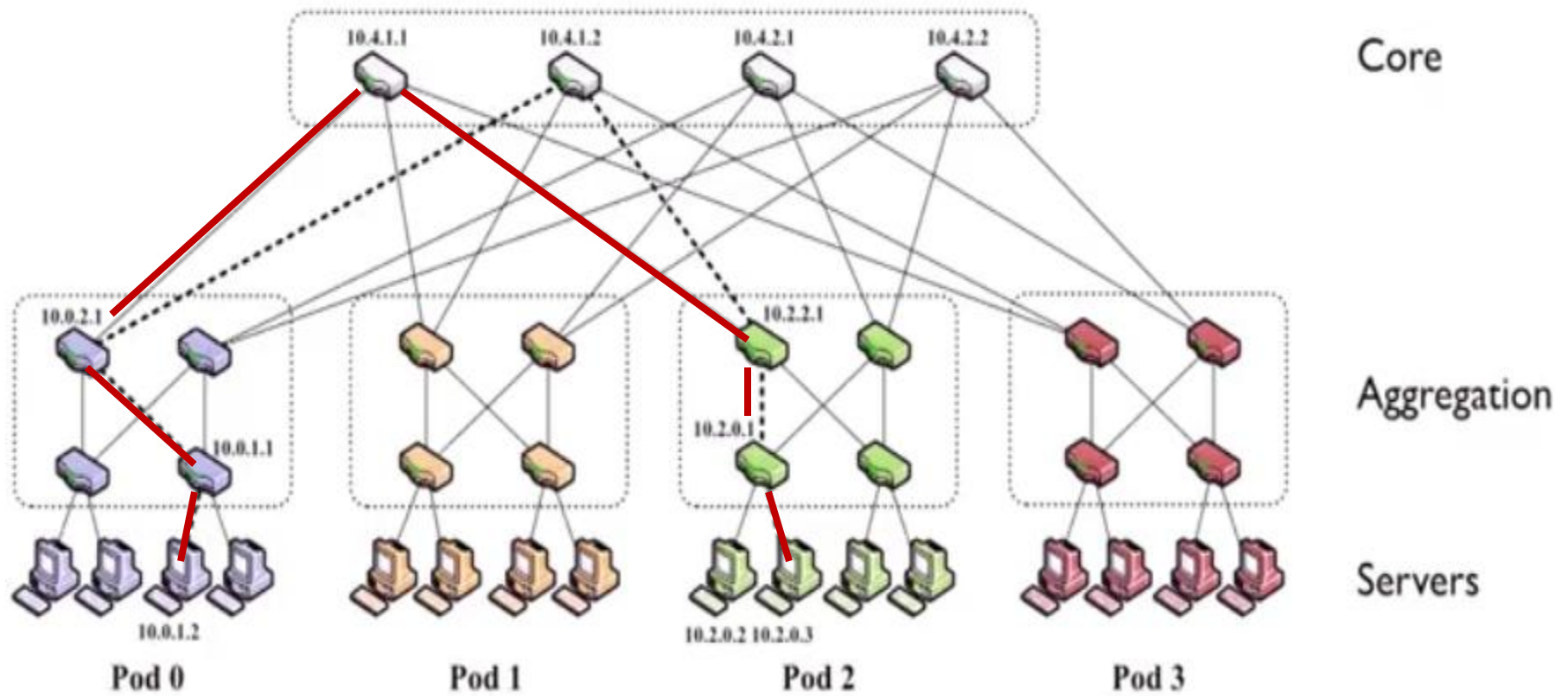
Another Solution: Clos networks

- Connects any number of input to any number of outputs
- Consists of small size switches



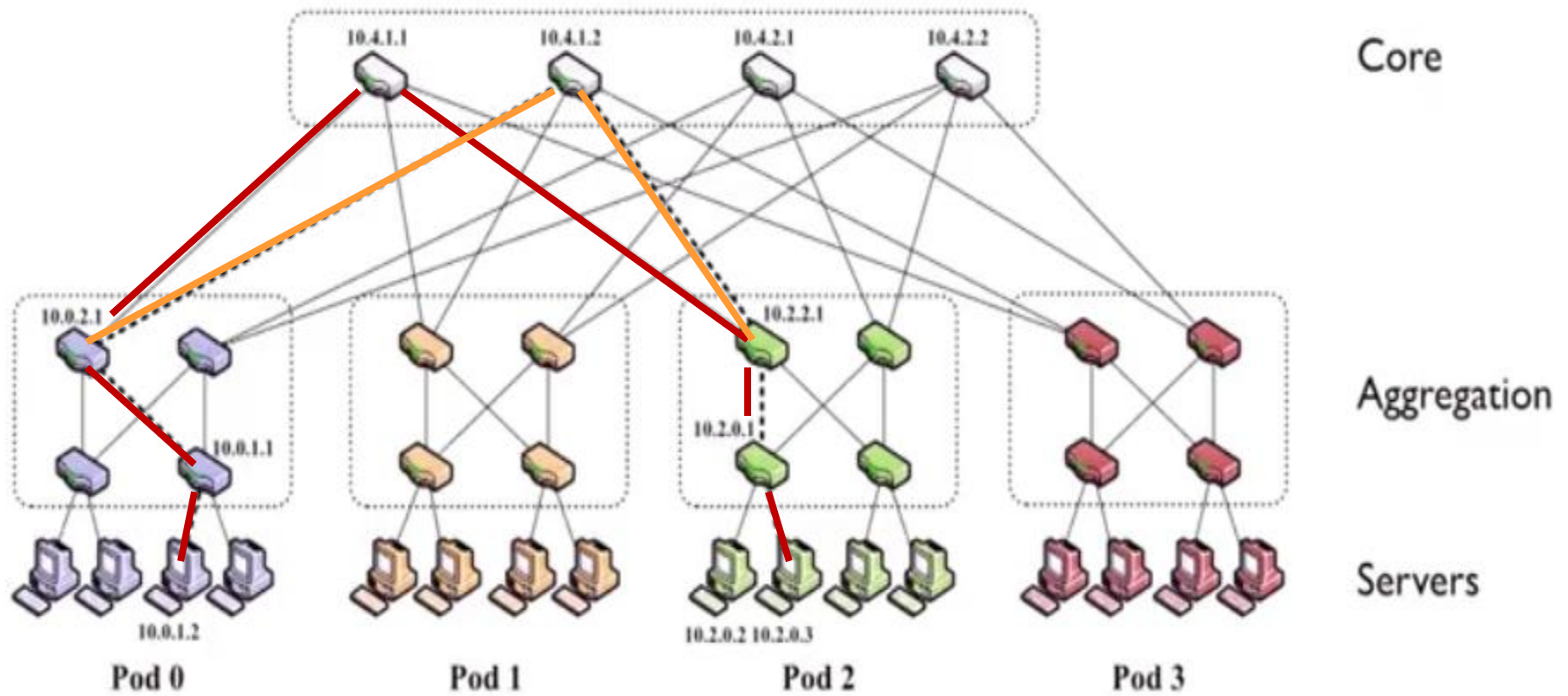
Physical Structure

Another Solution: Clos networks



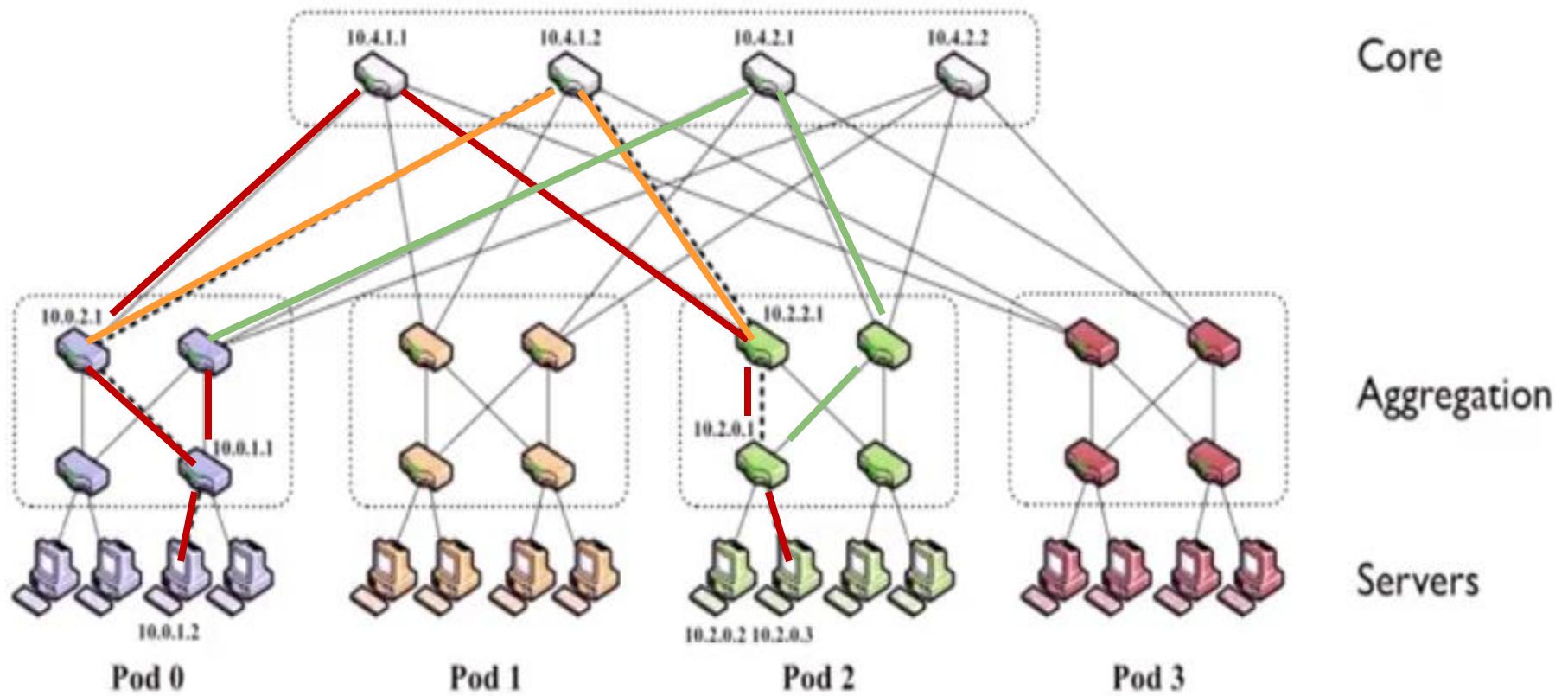
Physical Structure

Another Solution: Clos networks



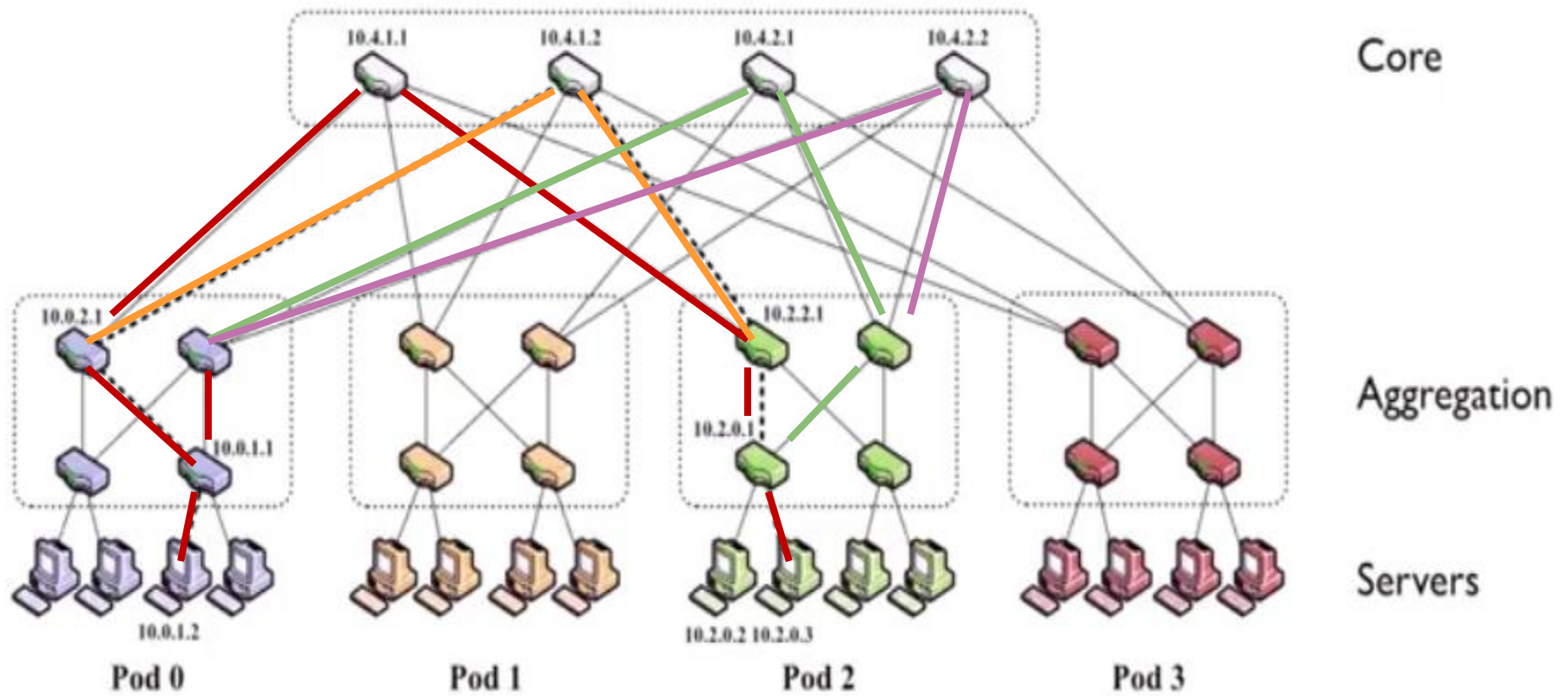
Physical Structure

Another Solution: Clos networks

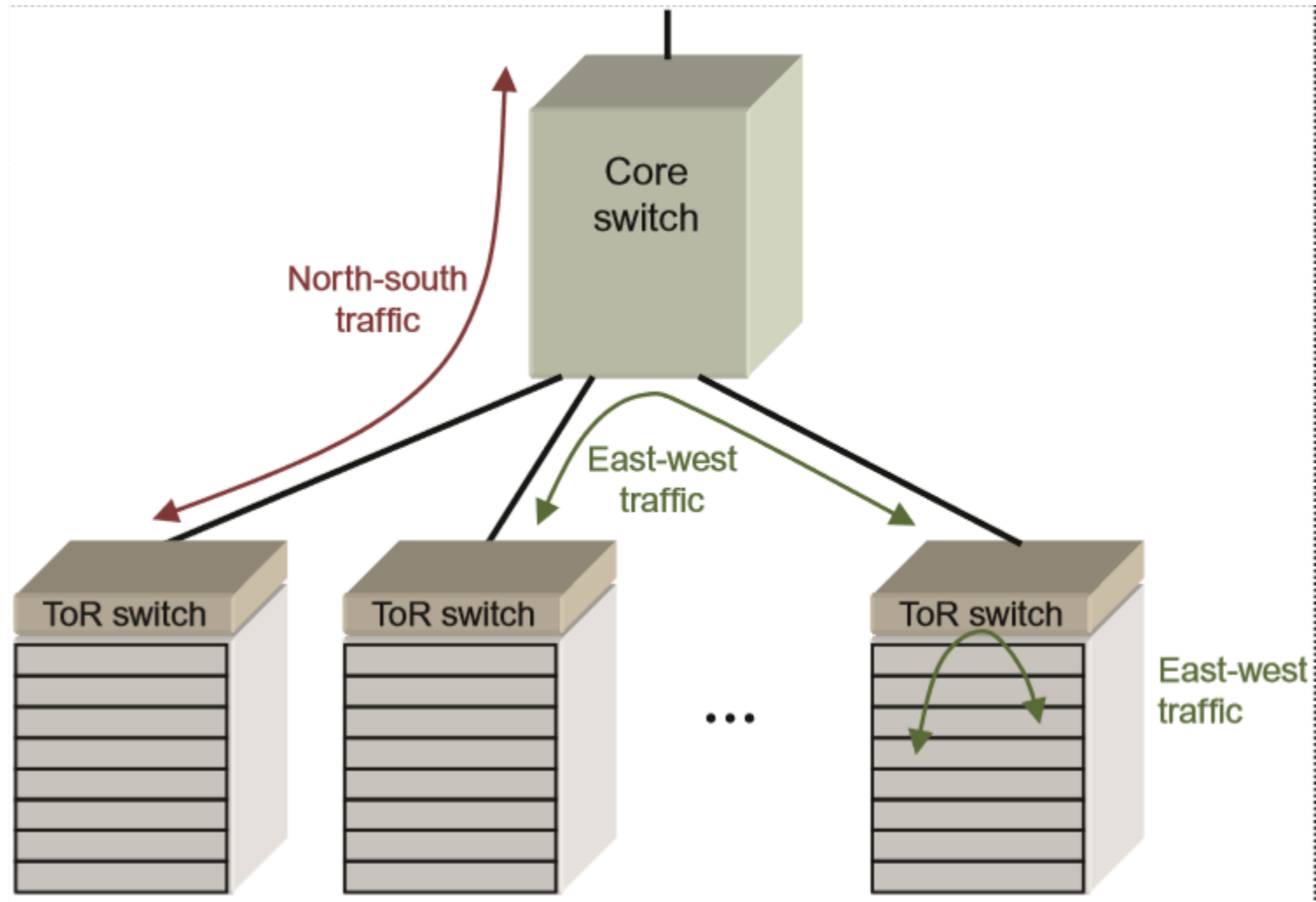


Physical Structure

Another Solution: Clos networks



Physical Structure





Questions?